Project Assignment #3

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1 Question 10

a) The `%.02f` option helps format the `statefips` variable by instructing Stata to format it in a specific manner. In this case, we tell Stata to format it with a leading zero at the beginning of the number.

b) The assertion checks two things at the same time. It first checks whether the unemployment variable takes on the values 1, 2, or 3, and then checks whether the number of categories equals 3. In other words, it is verifying that all observations fall into these three categories.

c) This is because it is a left join. Instead of creating new observations for unmatched values in the using dataset, we tell Stata to only keep the observations that successfully merge with the master dataset.

d) If we wanted to merge based on the year when individuals were 18 instead of 16, we would adjust the formula for calculating that year accordingly. Since each person has a state FIPS and a calculated year attached to them, we would merge on that updated year variable. Essentially, the year variable is what allows us to match individuals to the correct year in the master file.

2 Question 11

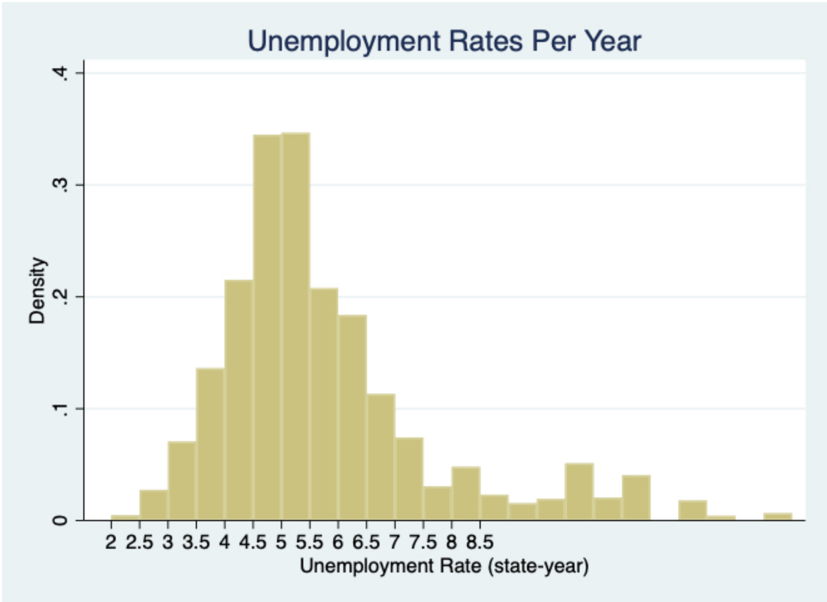

Figure 1: Histogram

3 Question 12

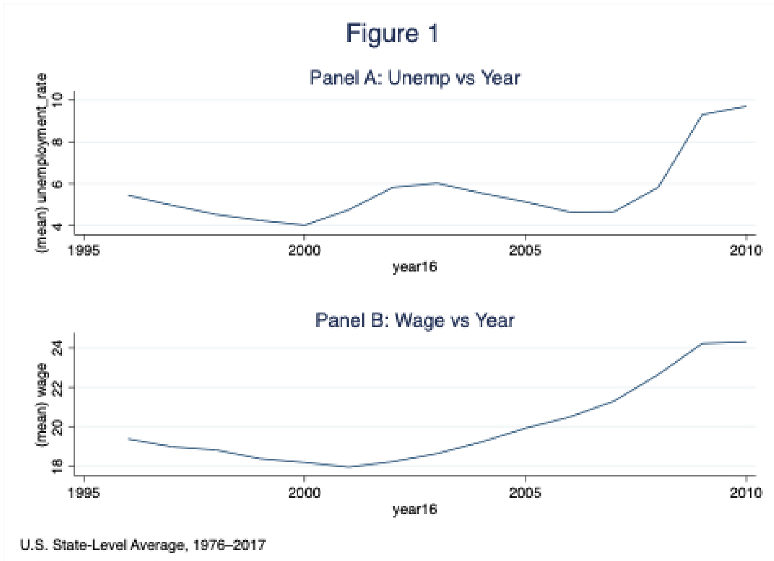

Figure 2: Figure 1

4 Question 13

Econ5813pa3-Table1

Table 1: Wage regressions				
="	(1)	(2)	(3)	(4)
="	wage	wage	wage	schoolyr
4% to 5.9%	0.841	1.463***	0.149	0.013
="	(0.549)	(0.341)	(0.132)	(0.015)
6% or more	3.296***	3.330***	-0.050	0.000
="	(0.560)	(0.412)	(0.264)	(0.024)
26	1.510***	1.670***	1.778***	0.141***
="	(0.060)	(0.082)	(0.081)	(0.008)
27	3.055***	3.365***	3.571***	0.230***
="	(0.122)	(0.175)	(0.167)	(0.012)
28	4.639***	5.071***	5.376***	0.309***
="	(0.220)	(0.312)	(0.275)	(0.013)
29	6.341***	6.907***	7.350***	0.387***
="	(0.291)	(0.429)	(0.374)	(0.017)
30	7.487***	8.321***	9.078***	0.407***
="	(0.304)	(0.520)	(0.447)	(0.016)
Hispanic	-5.510***	-5.849***	-5.921***	-1.813***
="	(0.880)	(0.827)	(0.832)	(0.113)
Black	-4.913***	-4.868***	-4.847***	-0.852***
="	(0.314)	(0.221)	(0.215)	(0.034)
Asian	6.061***	5.198***	5.076***	0.925***
="	(0.584)	(0.425)	(0.418)	(0.095)
Indian	-4.738***	-4.542***	-4.640***	-1.165***
="	(0.537)	(0.640)	(0.631)	(0.134)
Other race	-0.157	-0.633***	-1.058***	-0.164***
="	(0.162)	(0.120)	(0.130)	(0.030)
female	-2.120***	-2.159***	-2.141***	0.589***
="	(0.111)	(0.111)	(0.109)	(0.013)
State population (in tens of thousands)	0.001***	0.022***	0.003	0.001**
="	(0.000)	(0.004)	(0.003)	(0.000)
Constant	15.107***	2.782	8.241***	12.566***
="	(0.603)	(1.908)	(1.217)	(0.100)
State fixed effects	No	Yes	Yes	Yes
Year fixed effects	No	No	Yes	Yes
Mean of dependent variable	20.3	20.3	20.3	13.9
R-squared	.0761	.102	.113	.13
Sample size	1336498	1336498	1336498	1336498
Standard errors in parentheses				
="* p<0.05	** p<0.01	*** p<0.001"		

Figure 3: Regression Table

5 Question 14

One mechanism is that if there is less money in the household due to parental unemployment, the student may be forced to leave school or work, leading to lower educational attainment.

Another mechanism is the opposite: higher unemployment may encourage them to stay in school longer and accumulate more human capital, since outside job opportunities are limited. This would predict a positive sign. This is reflected in the low, medium, and high unemployment categories we use.

6 Question 15

$$wage_{ist} = \beta_1 \mathbf{1}\{unemp\_cat3_i = 1\} + \beta_2 \mathbf{1}\{unemp\_cat3_i = 2\} + \beta_3 \mathbf{1}\{unemp\_cat3_i = 3\} \quad + \gamma_s + \delta_t + X_i' \theta + \varepsilon_{ist}, \tag{1}$$

We find that none of the unemployment category variables are statistically significant. This suggests that these mechanisms may not strongly affect the outcome variable in this dataset. If any mechanism exists, the results are too imprecise to detect it reliably.

7 Question 16

Model 1 does not account for inherent state differences that do not change over time, such as voter composition or long-standing policy differences. Since Model 2 includes state fixed effects, it removes this source of bias. Model 3 is better than Model 2 because it also accounts for time variation—for example, differences in national economic conditions across years.

However, Model 3 could also be biased if something that drives unemployment varies by year and is correlated with wages. That would induce bias because the running variable is not independent of year. Therefore, Model 2 may provide a better estimate if Model 3 suffers from this source of bias.

8 Question 17

It appears that schooling is only somewhat related to the outcome, as both coefficients are positive but statistically insignificant. This may be because truck drivers often do not need education beyond a high school diploma, so additional schooling does not strongly influence their wages.

9 Question 18

You would not add school year because it would be perfectly correlated with our year variables; school year is just a linear transformation of year, which would create perfect multicollinearity.

10 Question 19

Using birth year instead of `year16` produces the exact same regression results. This makes sense because the two variables differ only by a constant shift of 16 years. We simply shift the x-axis by 16 units, so the underlying variation is identical.

11 Question 20

i) No, state population is not statistically significant in my regression. This is likely because state fixed effects already absorb much of the relevant information. Since our main variables of interest remain statistically insignificant regardless, I agree with the argument that we do not need to include population as a control. Including it does not change the results and does not contribute meaningful explanatory power.

ii) I would explain that unemployment likely does not have linear effects on wages. For example, a change from 1% to 3% unemployment may not be as impactful as a change from 6% to 7%, where job scarcity becomes more severe. Using categories allows the model to capture this nonlinearity. The continuous unemployment rate is statistically insignificant and points in the same direction as before. Splitting it into categories does not materially change the regression results, so both approaches tell the same story.